PTZ Control Configuration

Video & Streaming - PTZ Control Configuration

This guide covers the configuration of PTZ (Pan-Tilt-Zoom) control for cameras in GCXONE, including device-side setup, platform configuration, preset management, and automation.

Estimated time: 20–30 minutes per PTZ camera

Prerequisites

Before configuring PTZ control, ensure:

- PTZ camera is successfully added to GCXONE
- Camera hardware supports PTZ functionality
- Device is online and accessible
- Video streaming is working correctly
- Appropriate user permissions are available

Configuration Workflow

1. Device-Level PTZ Configuration
2. GCXONE PTZ Settings
3. Preset Configuration
4. Preset Tours
5. PTZ Automation
6. Verification

Part 1: Device-Level PTZ Configuration

Step 1: Enable PTZ on Camera Device

Navigation: Device Web Interface → PTZ Settings / Camera Control

1. Access the camera's web interface
2. Navigate to PTZ Settings or Camera Control
3. Enable PTZ Function:
    - Enable PTZ and select appropriate protocol such as ONVIF or manufacturer-specific
    - Verify PTZ Address — typically 1 or auto-detect
4. Configure Movement Limits if applicable:
    - Set minimum and maximum pan and tilt angles
    - Configure maximum movement speeds

5. Configure PTZ Parameters:
    - Idle Time: Time before camera returns to home position
    - Home Position: Define default position
6. Click Save or Apply

Note: Exact navigation varies by manufacturer. Refer to device documentation.

Step 2: Configure PTZ Protocol Settings

Navigation: Device Web Interface → Network → PTZ / Protocol Settings

1. Verify PTZ Protocol — enable ONVIF PTZ for compatibility or native manufacturer protocol if available
2. Verify PTZ control port — typically the same as the device HTTP port
3. Ensure the integration user has PTZ control permissions
4. Test PTZ from device interface — verify pan, tilt, zoom, and preset save/recall
5. Click Save

Part 2: GCXONE Platform PTZ Configuration

Step 3: Enable PTZ Control in GCXONE

Navigation: GCXONE → Devices → [Camera Name] → Camera Configuration → PTZ Settings

1. Navigate to Devices and select your PTZ camera
2. Click Camera Configuration and navigate to PTZ Settings tab
3. Enable PTZ Control and select protocol — Auto-detect or specific
4. Click Test PTZ Connection to verify communication
5. Configure PTZ Options:
    - Default Speed: Set default movement speed from 1–8
    - Invert Controls: Enable if needed for camera orientation
    - Relative Movement: Enable for incremental movement
6. Configure Advanced Settings:
    - Control Timeout: Time before control is released if multiple users
    - Enable Preset Support and Tour Support if applicable
7. Click Save Configuration

Step 4: Configure PTZ User Permissions

Navigation: GCXONE → Settings → User Management → [User] → Permissions

1. Select user or role to configure
2. Enable PTZ Control permissions:
    - View PTZ Controls: to see PTZ interface
    - Control PTZ: to operate camera
    - Manage Presets: to create and edit presets
    - Control Tours: if using tours

3. Configure PTZ Restrictions if needed:
    - Restrict to specific cameras
    - Limit PTZ control to specific times
4. Click Save Permissions

Part 3: Preset Configuration

Step 5: Create Preset Positions

Navigation: GCXONE → Live View → [PTZ Camera] → PTZ Controls → Presets

1. Navigate to Live View and select the PTZ camera
2. Open PTZ Controls panel
3. Use PTZ controls to position camera — pan, tilt, zoom, and focus as needed
4. Once positioned correctly:
    - Click Save Preset or the + button
    - Enter a descriptive preset name such as "Front Entrance" or "Parking Lot North"
    - Enter a preset number from 1–255
    - Click Save or OK
5. Verify preset appears in the list and test recalling it

Best practices for presets:

- Use descriptive names indicating location or purpose
- Number critical presets sequentially from 1–10 for quick access
- Test each preset after creation
- Update presets if camera position changes

Step 6: Organize and Manage Presets

Navigation: GCXONE → Devices → [Camera] → PTZ Settings → Preset Management

- Edit Preset: Click on preset name to update or reposition
- Reorder Presets: Drag and drop to reorder or renumber for logical organization
- Delete Preset: Select preset, click Delete, and confirm
- Export/Import Presets: Export for backup or import to duplicate on other cameras

Part 4: Preset Tours Configuration

Step 7: Create Preset Tours

Navigation: GCXONE → Devices → [Camera] → PTZ Settings → Tours

1. Click Create Tour or Add Tour
2. Configure Tour Settings:
    - Tour Name: such as "Perimeter Scan" or "Entrance Check"
    - Enable Repeat for continuous tours

3. Add Presets to Tour:
    - Select presets to include
    - Set Dwell Time — how long to stay at each preset
    - Set Movement Speed between presets
    - Reorder presets to define tour sequence
4. Configure Tour Options:
    - Start Position, End Behavior, and whether the tour is interruptible
5. Click Save Tour

Step 8: Configure Tour Automation

Navigation: GCXONE → Settings → Automation → PTZ Tours

1. Create Tour Schedule:
    - Select tour and set time of day and days of week
2. Configure Event-Triggered Tours:
    - Set trigger such as alarm type or time period
3. Set Tour Priorities — define which tours can interrupt others
4. Click Save Automation

Part 5: PTZ Automation Rules

Step 9: Configure Alarm-Triggered PTZ Movements

Navigation: GCXONE → Settings → Automation → PTZ Rules

1. Click Create Rule or Add Rule
2. Configure Rule Conditions:
    - Trigger: Select alarm type or event
    - Camera/Sensor: Select which sensor triggers the rule
    - Optional time restrictions
3. Configure PTZ Action:
    - Select target PTZ camera
    - Choose action: move to preset, start tour, or specific movement
4. Configure Rule Options:
    - Delay before action
    - Duration at preset
    - Return behavior after action
5. Click Save Rule

Example: When motion is detected at the Front Door sensor, move PTZ camera to Preset 1 (Front Entrance) for 60 seconds, then return to home position.

Part 6: Verification Checklist

Basic PTZ Control

- Pan left/right works correctly
- Tilt up/down works correctly
- Zoom in/out works correctly
- Movement speed is appropriate and controls are responsive

Preset Functionality

- Can save new presets and recall them to correct positions
- Can edit and delete presets
- Presets are organized logically with descriptive names

Advanced Features

- Preset tours work and visit all presets in sequence
- Tours can be interrupted and resumed
- Alarm-triggered PTZ movements work correctly

Performance

- PTZ response time is acceptable with no lag
- Movement is smooth and reaches preset positions accurately

Advanced Configuration

Multi-Camera PTZ Coordination

- Use consistent preset numbering across cameras
- Configure synchronized tours for multiple cameras
- Group related presets across cameras for quick access

PTZ Performance Optimization

- Use Local Mode for better PTZ responsiveness
- Adjust speed based on use case — faster for large movements, slower for precision
- Set appropriate control timeout to prevent conflicts between users

Quick Troubleshooting

- PTZ not working: Verify PTZ is enabled on device, check protocol settings, verify user permissions
- Presets not saving: Check preset count limit, verify permissions, check device storage
- Slow response: Check network latency, use Local Mode if available
- Movement inaccurate: Calibrate camera, check for mechanical issues, verify preset positions